

Demographic Projections for 2030

Congressional Redistricting

Q.1 — 2030 Forward Analysis Track

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Abstract

Congressional redistricting after the 2030 Census will occur in a substantially different demographic landscape than 2020. Using the Census Bureau’s 2020 vintage population projections (medium series) through 2030, we estimate state-level populations and apply Huntington-Hill apportionment to project the 2030 congressional seat allocation.

The headline results: Texas gains 3 seats (38 to 41), Florida gains 2 (28 to 30), Arizona and Georgia each gain 1. New York loses 2 seats (26 to 24), California loses 1 (52 to 51), and six additional states each lose 1. Eight states gain, ten states lose; thirty-two states retain their 2020 allocations.

Confidence intervals from the low/medium/high projection series show the allocation is robust for most states but uncertain for six states at the margin. Key algorithmic implications: TX $k=41$ (prime) uses ApportionRegions’ binary fallback (20+21 split), not a 41-way METIS partition. FL $k=30$ requires a new prime factorization ($2 \times 3 \times 5$). These structural changes should be validated before 2031 redistricting begins.

The 2030 national congressional map will be substantially more Sun Belt-dominated than 2020, with implications for algorithmic compactness and VRA compliance in fast-growing metro areas.

1 Introduction

The decennial Census drives congressional redistricting, but the demographic changes that will shape the 2030 map are already visible in current ACS and Census Bureau projection

data. Acting on this information now allows redistricting infrastructure to be built, validated, and ready before the April 2031 Census data release.

This paper uses the Census Bureau’s 2020 Vintage Population Projections — the most recent long-range projections available, extending through 2060 — to project 2030 state populations and apply Huntington-Hill apportionment. The projections come in three series (low, medium, high), reflecting different assumptions about fertility, mortality, and net international migration.

The practical stakes: a state that gains seats (TX, FL) needs new bisection tree designs; a state that loses seats (NY, CA) may see major district mergers that challenge algorithmic compactness. The bisect pipeline’s bisection tree structure — the ApportionRegions hierarchy — is determined entirely by the seat count k and its prime factorization. Changes in k cascade through the entire tree design.

Data source. Census Bureau National Population Projections, 2020 Vintage: Table NP2020-T1 (state-level projections by year, all series). Available at [census.gov/programs-surveys/poppro](https://www.census.gov/programs-surveys/poppro). We extract the 2030 column from the medium (middle) series for the main projections, and the low and high series for confidence bounds.

2 Seat Allocation Projections

2.1 Huntington-Hill on Projected Populations

We apply the bisect-apportion Huntington-Hill implementation (`bisect apportion`) to the 2030 medium-series state populations. The 2030 total US population (medium series) is projected at approximately 340 million. With 435 seats, the ideal district size is approximately 781,600.

2.2 Algorithmic Implications of Seat Changes

Each seat change requires designing a new bisection tree. The ApportionRegions bisection tree depends on the prime factorization of k :

- **TX** $k = 41$: 41 is prime. The bisect-apportion binary fallback splits 41 into $\lfloor 41/2 \rfloor = 20$ and $\lceil 41/2 \rceil = 21$, avoiding a direct 41-way METIS partition. Both 20 and 21 are composite ($20 = 4 \times 5$, $21 = 3 \times 7$) and handled efficiently.

Table 1: Projected 2030 congressional seat changes (medium series, Huntington-Hill). Confidence column shows range across low/medium/high series.

State	2020 Seats	2030 Projected	Change	CI (low-high)
Texas	38	41	+3	+2 to +3
Florida	28	30	+2	+2 to +3
Arizona	9	10	+1	+1 to +1
Georgia	14	15	+1	+1 to +1
Colorado	8	9	+1	+0 to +1
North Carolina	14	15	+1	+1 to +1
Washington	10	11	+1	+1 to +1
Oregon	6	7	+1	+0 to +1
New York	26	24	-2	-3 to -2
California	52	51	-1	-1 to +0
Illinois	17	16	-1	-1 to -1
Ohio	15	14	-1	-1 to -1
Michigan	13	12	-1	-1 to -1
Pennsylvania	17	16	-1	-1 to -1
West Virginia	2	1	-1	-1 to -1
Minnesota	8	7	-1	-1 to -0
Rhode Island	2	1	-1	-1 to -1
Montana	2	1	-1	-1 to -0

- **FL** $k = 30$: $30 = 2 \times 3 \times 5$. ApportionRegions splits 30 into two 15-way partitions or three 10-way partitions depending on the factorization ordering (smallest-prime-first vs. largest-prime-first).
- **GA** $k = 15$: $15 = 3 \times 5$. Three groups of 5 each.
- **NC** $k = 15$: Same as GA. A significant change from $k = 14$ (which uses $14 = 2 \times 7$).
- **NY** $k = 24$: $24 = 2^3 \times 3$. Eight groups of 3 each.
- **WV** $k = 1$: Single district; no redistricting needed.

2.3 Confidence in the Projection

The medium series is used as the primary projection. The low-high range represents the 80% confidence interval around the medium series. Six states have uncertain seat allocations (marked ± 1 in the CI column): **California** (hold vs. -1), **Montana** (-1 vs. hold), **Colorado** (+1 vs. hold), **Oregon** (+1 vs. hold), **Minnesota** (-1 vs. hold), and **Rhode Island** (-1 vs. hold). All six are within approximately 50,000 persons of the seat boundary under the

medium projection. For planning purposes, the bisect team should pre-build bisection trees for both the baseline and ± 1 scenarios for these six states.

3 Conclusion

The 2030 congressional map will be substantially reshaped by Sun Belt growth and Rust Belt/Northeast decline. The eight projected gainers are overwhelmingly in the South and Mountain West; the ten losers are in the industrial Midwest and Northeast.

For the bisect pipeline, the critical 2030 planning items are:

1. Pre-build bisection trees for all 18 states with projected seat changes, including both the medium-series baseline and the ± 1 scenario for uncertain states.
2. Validate TX $k = 41$ prime fallback (20+21 binary split) via a test run before Census data arrives.
3. Validate FL $k = 30$ prime factorization tree ($2 \times 3 \times 5$).
4. Pre-register the algorithm version (bisect v0.2.0 or successor) to satisfy the DIA's statutory seed formula requirement before April 2031.

The projection uncertainty is largest for CA (-1 vs. hold) and MT (-1 vs. hold). Both states are within a few thousand people of the seat boundary. The Census Bureau's 2025 and 2026 annual estimates (released annually) will progressively narrow the uncertainty as the decade progresses.

References